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Decision Table

Five worked examples of condition-rule-action logic, with derived test cases for each.

1. Restaurant Table Reservation

Governs whether a diner's table request is confirmed based on availability, party details, and restaurant policy checks.

Conditions vs. Rules

Condition	R1	R2	R3	R4	R5
C1: Table Available	Y	N	Y	Y	Y
C2: Party Size Within Limit	Y	-	N	Y	Y
C3: Valid Contact Number	Y	-	-	N	Y
C4: Booking Window Open	Y	-	-	-	N
C5: Special Event Block Clear	Y	-	-	-	-

Y = Yes, N = No, - = Not applicable / not checked for this rule.

Resulting Actions

Action	R1	R2	R3	R4	R5
A1: Table Reserved	X				
A2: Show "No Tables Available"		X			
A3: Show "Party Size Exceeds Limit"			X		
A4: Show "Contact Number Invalid"				X	
A5: Show "Booking Window Closed"					X

An X marks the action that fires for that rule; all other actions stay off.

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC1_01	Table available, party within limit, contact valid, booking window open, no event block	Table reserved	R1

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC1_02	No table available for requested time	Show 'No Tables Available'	R2
TC1_03	Table available but party size exceeds limit	Show 'Party Size Exceeds Limit'	R3
TC1_04	Table and party size fine, contact number invalid	Show 'Contact Number Invalid'	R4
TC1_05	Booking window closed for the day	Show 'Booking Window Closed'	R5

Notes on Each Test Case

- **TC1_01** – checks that the result is “Table reserved” when the R1 conditions are met.
- **TC1_02** – checks that the result is “Show 'No Tables Available'” when the R2 conditions are met.
- **TC1_03** – checks that the result is “Show 'Party Size Exceeds Limit'” when the R3 conditions are met.
- **TC1_04** – checks that the result is “Show 'Contact Number Invalid'” when the R4 conditions are met.
- **TC1_05** – checks that the result is “Show 'Booking Window Closed'” when the R5 conditions are met.

2. Movie Ticket Booking

Determines the outcome of a cinema seat-booking attempt based on show availability, age rating, and payment checks.

Conditions vs. Rules

Condition	R1	R2	R3	R4	R5
C1: Seats Available	Y	N	Y	Y	Y
C2: Age Rating Requirement Met	Y	-	N	Y	Y
C3: Payment Successful	Y	-	-	N	Y
C4: Coupon Code Valid	Y	-	-	-	-
C5: Show Time Not Passed	Y	-	-	-	N

Y = Yes, N = No, - = Not applicable / not checked for this rule.

Resulting Actions

Action	R1	R2	R3	R4	R5
A1: Ticket Confirmed	X				
A2: Show "Sold Out"		X			
A3: Show "Age Rating Restricted"			X		
A4: Show "Payment Declined"				X	
A5: Show "Show Time Passed"					X

An X marks the action that fires for that rule; all other actions stay off.

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC2_01	All conditions satisfied	Ticket confirmed	R1
TC2_02	Show is sold out	Show 'Sold Out'	R2
TC2_03	Viewer does not meet age rating	Show 'Age Rating Restricted'	R3
TC2_04	Payment declined at checkout	Show 'Payment Declined'	R4
TC2_05	Show time already passed	Show 'Show Time Passed'	R5

Notes on Each Test Case

- **TC2_01** – checks that the result is “Ticket confirmed” when the R1 conditions are met.
- **TC2_02** – checks that the result is “Show 'Sold Out'” when the R2 conditions are met.
- **TC2_03** – checks that the result is “Show 'Age Rating Restricted'” when the R3 conditions are met.
- **TC2_04** – checks that the result is “Show 'Payment Declined'” when the R4 conditions are met.
- **TC2_05** – checks that the result is “Show 'Show Time Passed'” when the R5 conditions are met.

3. Car Rental Booking

Controls whether a vehicle rental request is approved based on fleet availability, licence status, and driver eligibility.

Conditions vs. Rules

Condition	R1	R2	R3	R4	R5
C1: Vehicle Available	Y	N	Y	Y	Y
C2: Driving Licence Valid	Y	-	N	Y	Y
C3: Payment Authorized	Y	-	-	N	Y
C4: Minimum Age Met	Y	-	-	-	N
C5: Pickup Slot Confirmed	Y	-	-	-	-

Y = Yes, N = No, - = Not applicable / not checked for this rule.

Resulting Actions

Action	R1	R2	R3	R4	R5
A1: Rental Confirmed	X				
A2: Show "No Vehicles Available"		X			
A3: Show "Licence Invalid"			X		
A4: Show "Payment Not Authorized"				X	
A5: Show "Below Minimum Age"					X

An X marks the action that fires for that rule; all other actions stay off.

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC3_01	All conditions satisfied	Rental confirmed	R1
TC3_02	No vehicles left in the requested class	Show 'No Vehicles Available'	R2
TC3_03	Driving licence expired or invalid	Show 'Licence Invalid'	R3
TC3_04	Payment authorization fails	Show 'Payment Not Authorized'	R4
TC3_05	Driver below minimum age requirement	Show 'Below Minimum Age'	R5

Notes on Each Test Case

- **TC3_01** – checks that the result is “Rental confirmed” when the R1 conditions are met.
- **TC3_02** – checks that the result is “Show 'No Vehicles Available'” when the R2 conditions are met.
- **TC3_03** – checks that the result is “Show 'Licence Invalid'” when the R3 conditions are met.
- **TC3_04** – checks that the result is “Show 'Payment Not Authorized'” when the R4 conditions are met.

- **TC3_05** – checks that the result is “Show 'Below Minimum Age'” when the R5 conditions are met.

4. Hotel Check-In

Defines how the front-desk system responds to a guest check-in attempt based on booking status and verification checks.

Conditions vs. Rules

Condition	R1	R2	R3	R4	R5
C1: Room Ready	Y	N	Y	Y	Y
C2: Booking Confirmed	Y	-	N	Y	Y
C3: ID Verified	Y	-	-	N	Y
C4: Payment Method On File	Y	-	-	-	Y
C5: Check-in Time Reached	Y	-	-	-	N

Y = Yes, N = No, - = Not applicable / not checked for this rule.

Resulting Actions

Action	R1	R2	R3	R4	R5
A1: Check-In Completed	X				
A2: Show "Room Not Ready"		X			
A3: Show "Booking Not Found"			X		
A4: Show "Identity Verification Failed"				X	
A5: Show "Too Early to Check In"					X

An X marks the action that fires for that rule; all other actions stay off.

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC4_01	All conditions satisfied	Check-in completed	R1
TC4_02	Room not yet ready for occupancy	Show 'Room Not Ready'	R2
TC4_03	No matching booking found	Show 'Booking Not Found'	R3
TC4_04	Guest ID could not be verified	Show 'Identity Verification Failed'	R4
TC4_05	Guest arrives before check-in time	Show 'Too Early to Check In'	R5

Notes on Each Test Case

- **TC4_01** – checks that the result is “Check-in completed” when the R1 conditions are met.
- **TC4_02** – checks that the result is “Show 'Room Not Ready'” when the R2 conditions are met.
- **TC4_03** – checks that the result is “Show 'Booking Not Found'” when the R3 conditions are met.
- **TC4_04** – checks that the result is “Show 'Identity Verification Failed'” when the R4 conditions are met.

- **TC4_05** – checks that the result is “Show 'Too Early to Check In'” when the R5 conditions are met.

5. Online Course Enrollment

Governs whether a learner's enrollment request into an online course succeeds, based on seat capacity and prerequisite checks.

Conditions vs. Rules

Condition	R1	R2	R3	R4	R5
C1: Seat Capacity Available	Y	N	Y	Y	Y
C2: Prerequisites Completed	Y	-	N	Y	Y
C3: Payment Successful	Y	-	-	N	Y
C4: Enrollment Window Open	Y	-	-	-	N
C5: Profile Fully Verified	Y	-	-	-	-

Y = Yes, N = No, - = Not applicable / not checked for this rule.

Resulting Actions

Action	R1	R2	R3	R4	R5
A1: Enrollment Confirmed	X				
A2: Show "Course Full"		X			
A3: Show "Prerequisites Not Met"			X		
A4: Show "Payment Failed"				X	
A5: Show "Enrollment Closed"					X

An X marks the action that fires for that rule; all other actions stay off.

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC5_01	All conditions satisfied	Enrollment confirmed	R1
TC5_02	Course has reached seat capacity	Show 'Course Full'	R2
TC5_03	Learner has not completed prerequisites	Show 'Prerequisites Not Met'	R3
TC5_04	Payment could not be processed	Show 'Payment Failed'	R4
TC5_05	Enrollment window has closed	Show 'Enrollment Closed'	R5

Notes on Each Test Case

- **TC5_01** – checks that the result is “Enrollment confirmed” when the R1 conditions are met.
- **TC5_02** – checks that the result is “Show 'Course Full'” when the R2 conditions are met.
- **TC5_03** – checks that the result is “Show 'Prerequisites Not Met'” when the R3 conditions are met.
- **TC5_04** – checks that the result is “Show 'Payment Failed'” when the R4 conditions are met.
- **TC5_05** – checks that the result is “Show 'Enrollment Closed'” when the R5 conditions are met.